

1. Implement the numpy basics

<https://numpy.org/doc/stable/user/basics.html>

- Numpy Vs Python's built-in list data structure
 - Memory Consumption: ndarray vs list
 - Time Comparison between Python Lists and Numpy Arrays
- Numpy's array creating functions (eg. `.zeros`, `.ones`, `.full`, `.linspace`, and so on)
- Different standard numpy data types (eg. `bool_`, `int_`, `intp`, `int8`, `int16`, `float`, `complex_`, `complex64`, and so on)

2. Dataframes with Pandas:

<https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.html>

- Dataframes
Obtaining Basic Information About DataFrame, Selection and Indexing, Index Basics
- Filtering
filtering with selection brackets, `isin()`, `between()`, `contains()`, `query()`, `loc[]`, `pandas filter()`
- Useful Methods
`apply()`, `describe()`, `sort_values()`, `corr()`, `idxmin` and `idxmax`, `value_counts`, `replace`, `unique` and `nunique`, `map`, `uplicated` and `drop_duplicates`, `between`
- Handling missing data
`isnull()`, `dropna()`, `fillna()`

3. Data Visualization

The task is to learn the following with Matplotlib (optionally seaborn):

<https://matplotlib.org/stable/users/explain/figure/index.html>

- Understanding the Figure Object
- Implementing Figures and Axes
- Figure Parameters
- Subplots Functionality
- Styling - Legends
- Styling - Colors and Styles

You can utilize the data analysis from pandas for plotting visualization.

4. FastAPI

<https://fastapi.tiangolo.com/tutorial/>

Implement the following API endpoints for your resource:

- Create: Allow users to add a new resource. This should be accessible via a POST request.
- Read: Allow users to retrieve a list of all resources or a specific resource by its identifier (e.g., via GET requests).
- Update: Allow users to update an existing resource using its identifier. This should be accessible via a PUT request.
- Delete: Allow users to delete a resource by its identifier. This should be accessible via a DELETE request.

